ASSESSING IMPACTS OF INSTITUTIONAL FOOD PROCUREMENT

If you have access to Jupyter Notebooks python-based software that can link to an OpenAI account, you might first try downloading from this folder to your instance of Jupyter at least these files: 1) FoodImpacts_v10d.ipynb 2) usfoods.csv (purchases by Yale from USFoods.com); 3) factors.csv; and 4) foodcategories.json. Those are among the files in _AllFilesForDownload.zip.

Try running the FoodImpacts_v10d.ipynb Notebook. That should complete in fewer than five minutes at an OpenAI cost of less than $.50, and assess the impact of the food procurement specified in usfoods.csv (1700+ food products), with two output files: impacts_lbs_usfoods.csv and impacts_metric_usfoods.csv. The first of those should be similar to this below, but with 60+ additional rows encompassing further food categories (see a sample in this folder).

CATEGORY	(weight/lbs)	(calories)	(lbs CO2e)	(lbs carbon opp costs)	(direct animal lives)	(total animal lives)	(water gallons)	(land use acres)	(lbs PO4-eq eutrophication)
additive	21016	0	0	0	0	0	0	0	0
almond milk	900	134713	625	612	0	0	40009	0.243	1.34996
anchovies	34.9	17868	135	0	374.8	404.8	0	0	0
animal fats	42	130570	293	783	2.7	7.7	7556	0.342	6.22144
apples	6798.8	989929	2434	6391	0	0	146642	2.267	9.85827
bananas	64.4	17596	46	73	0	0	888	0.068	0.21201
barley (beer)	7	1327	7	38	0	0	14	0.005	0.01631
beans and pulses (•	58773	92320110	98621	589493	0	0	3070553	414.72	1003.84216
beef	41449.7	19008123	1713781	8358340	47	51.3	7832603	2855.402	12758.22997
berries	8789.2	1192023	14546	8262	0	0	442333	3.371	53.78966
bivalves	783.3	55073	244	0	0	0	0	0	0
butter	9796.3	31860159	109170	412719	0.4	1	4633213	74.463	656.25481
buttermilk	7683.5	2160812	17050	48713	0.1	0.1	592001	8.795	83.82699
cabbages and other•	3529.3	360193	2195	2506	0	0	48211	4.432	17.68169
cassava and other •	3065.2	814760	1217	5119	0	0	0	0.868	2.11502
chicken	148245.9	96359614	652282	2179214	42166.6	44689	11724090	929.605	7219.57387

The output of impacts_lbs_usfoods.csv will show:
1) name of food category
2) the total weight of food from the usfoods.csv file in the particular food category
3) total calories (kcal) for the food category
4) the mass in lbs of CO2 equivalent emissions
5) the carbon opportunity costs of the land use
6) number of animals consumed directly
7) number of lives cost, both directly in respect to animals taken to slaughter, as well as indirect (e.g., mortality of siblings that died too sick or young to be used, and animals used as feed.
8) "blue" gallons of irrigation, drinking, pond, plus processing water withdrawn to produce the product.
9) the sq ft of land use for production of that amount of food in the particular category
10) estimated eutrophication effects in phosphate equivalent lbs weights
While impacts_metric_usfoods.csv will show similar output, but with metric measurements.

Principal sources for the food conversion factors used thus far have been 1) Poore & Nemecek (2018), retrieved directly and as used in WRI's Cool Food Pledge calculator 2) Faunalytics animal lives calculations (with this project's 2025 revisions) 3) Feitz et al for allocation of impacts to various milk products, relying on proportion of milk solids 4) Kyttä, V., Roitto, M., Astaptsev, A. et al. (2022) re: beef/milk allocation factors; 5) Gephart et al (2021) for more refined analysis of aquatic foods; and 6) Welfare Footprint Project for aquaculture pre-harvest mortality data.

OUTLINE OF HOW THIS WORKS

In the folder with this documentation is a “usfoods.csv” food procurement worksheet reflecting purchases by Yale U. during 2020-2021 from a particular vendor. That’s one page of the spreadsheet of purchases that Yale submitted as part of the AASHE/STARS program. That usfoods.csv spreadsheet includes, for example, on row 1624, a purchase of 21.8 pounds of:

SOUP, CHICKEN NOODLE RTU LOW SODIUM

The open source, freely available, software we are developing calls on OpenAI to define the ingredients, and what percent of the overall weight they constitute, producing results like:

A	B	C	D	E	F	G	H
1624	SOUP CHICKEN NOODLE RTU LOW SODIUM	water	water	21.8	lbs	50	10.9
1624	SOUP CHICKEN NOODLE RTU LOW SODIUM	chicken broth	liquids	21.8	lbs	20	4.36
1624	SOUP CHICKEN NOODLE RTU LOW SODIUM	cooked chicken meat	chicken	21.8	lbs	10	2.18
1624	SOUP CHICKEN NOODLE RTU LOW SODIUM	noodles	grains	21.8	lbs	10	2.18
1624	SOUP CHICKEN NOODLE RTU LOW SODIUM	carrots	vegetables	21.8	lbs	5	1.09
1624	SOUP CHICKEN NOODLE RTU LOW SODIUM	celery	vegetables	21.8	lbs	4	0.87
1624	SOUP CHICKEN NOODLE RTU LOW SODIUM	onions	vegetables	21.8	lbs	1	0.22

where column C shows the predicted ingredients, D categorizes the ingredients, E/F show the product’s total weight, G is the estimated percentage of total weight of each ingredient and H, the predicted weight of each ingredient.

By such a process, the 1780 product items in the worksheet are separated into more than 3800 constituent ingredients (more than 500 variations) with their individual weights estimated. Those each are then assigned by OpenAI to one of these categories: animal fats, beef, chicken, duck, eggs, goat, lamb/mutton, pork, turkey, unidentified meat, veal, dairy milk, butter, buttermilk, dairy cheese, concentrated milk, cream, ghee, ice cream, lactose powder, low fat yogurt, milk powder, skim milk, whey powder, yogurt, almond milk, apples, bananas, barley (beer), beans and pulses (dried), berries, cabbages and other brassicas (broccoli), cassava and other roots, citrus fruit, cocoa, coffee, corn (maize), fruits, grains/cereals (except rice), legumes, oatmilk, oats (oatmeal), olives (oil), onions and leeks, other vegetables, palm (oil), peanuts/groundnuts, peas, potatoes, rapeseed/canola (oil), rice, rice milk, root vegetables, roots and tubers, soymilk, soybeans (oil), soybeans/tofu, stimulants & spices, sugars and sweeteners, sunflower (oil), tomatoes, tree nuts and seeds, vegetable oils, wheat/rye (bread/pasta/baked goods), wine grapes (wine), non-dairy cheese or yogurt, water, unknown, flavoring, additive, seaweed (dried), seaweed (fresh), bivalves, anchovies, bass, bass (wild-caught), bass (farmed), carp, silver or bighead carp, catfish, catfish (wild-caught), catfish (farmed), cod, congers, crab, fish (finfish), flounder, haddock, hake, halibut, herring, jacks, lobster, milkfish, milkfish (wild-caught), milkfish (farmed), mullets, redfish, redfish (wild-caught), redfish (farmed), salmon, salmon (wild-caught), salmon (farmed), sardines, sauries, shrimp, shrimp (wild-caught), shrimp (farmed), smelt, sole, squid or cuttlefish or octopus, tilapia, tilapia (wild-caught), tilapia (farmed), trout, trout (wild-caught), trout (farmed), tuna, tuna (bluefin tuna).

Next, the total weight purchased of each such category can be translated to impacts such as the number of affected animals’ lives, the lbs of CO2 equivalent greenhouse gasses, and gallons of water used in production. Those use the file “factors.csv”, which one could update or alter consistent with institutional goals and best available scientific understanding. The factors.ods file is not used in the process, but exists to help show how many of the elements of factors.csv were derived. Additional detailed explanation of that process is available in the factor_sources file.

BACKGROUND

Systems exist to help institutions measure the GHG impacts of their food purchases. See, for example the World Resource Institute's Cool Food Pledge and SIMAP (the Sustainability Indicator Management and Analysis Platform) developed by the University of New Hampshire's Sustainability Institute. The latter benefits from Rebecca Grekin's work categorizing foods at TASTE Food: The Automated Scope 3 Tool for Tracking Emissions from Food. The work to develop that was detailed in her thesis while a graduate student in the Energy Sciences Engineering Department at Stanford University.

Additionally, Faunalytics has done considerable work assessing the impacts of institutional food purchasing. See for example, The Impact of Replacing Animal Products and the Faunalytics "Number of Animal Lives Calculator", a copy of which can be found in the same folder as this documentation. Our project has developed an updated/revised version of that calculator as reflected in our factors.csv file which uses both a Faunalytics-style direct and also "direct + indirect" calculations methodology, the latter encompasses animals that die of other causes before slaughter, adjusts the number for loss of product between the farm and point of sales, and adds estimated "feed" animals' deaths (e.g., fish fed to pigs and chickens).

The Association for Sustainability in Higher Education (AASHE) has developed a "transparent, self-reporting framework for colleges and universities to measure their sustainability performance". It is known as The Sustainability Tracking, Assessment & Rating System (STARS).

It includes credits related to participating institutions': curriculum, research, campus engagement, public engagement, air & climate, buildings and grounds, energy use, food & dining, purchasing, transportation, waste, water, coordination & planning, diversity & affordability, investment & finance, wellbeing & work, innovation & leadership.

The food & dining food procurement portion of the standards largely awards credits consistent with the proportion of plant-based foods in purchases, and for reliance on food providers that are certified as having sustainable practices, including for environmental protection and animal welfare.

As an example, Yale submitted to STARS an impressively well-documented accounting of its food purchases that can be found by AASHE members at:
https://reports.aashe.org/institutions/yale-university-ct/report/2022-06-29/OP/food-dining/OP-7/ via the link to "FY21_Food_and_Beverage_Purchasing_Inventory.xlsx"

That includes 7 worksheets, one each from these vendors or distributors 1) Baldor Specialty Foods 2) Jordan Paige 3) J Polep 4) FreshPoint 5) US Foods 6) North Coast Seafood and 7) Jamix (combines purchases from multiple vendors).

The format of each of those is different, so each needs to be tweaked a little bit. Typically the process involves making a copy of the file and ensuring there is a single header row of column labels including "product", "product_weight", and "unit" (all lower case) and adding a label in the first column to indicate the format prior to being interpreted by the software here.

Please see the final two pages of this document for a list of each of the files in this folder with brief description of their purposes.

ADJUSTING FOR DIFFERENT SPREADSHEET FORMATS

Save the procurement worksheet as a .csv file and then adjust the column header row to use names like product, product_weight, unit, qty.

type01

If the original worksheet is in a format like this, with quantity and units of measure in one column (e.g. like Jordan Paige, J Polep, North Coast Seafood & Jamix worksheets from Yale):

Product	Quantity (base unit)
Harissa Paste (Tunisian) 40z EA JAR	80.000 OZ
PASTE MISO RED 2.2#	4.400 LB
PASTE HARISSA	84.000 lb
SPICE MASALA GARAM BULK	7.875 lb

then ensure the product column is named "product" and the "Quantity (base unit)" or similar column renamed to "product_weight". And IMPORTANT, somewhere in the first column (e.g., in the row after the last product) include "type01" (lower case, no space).

The third cell of the code will break the "product_weight" into two columns, e.g., separating the "80.000" and "OZ" into separate columns prior to futher processing.

When running the Notebook, for that Jordan Paige worksheet one may see several messages like:
"Skipping row 500 due to missing data on weight or unit.
Unit 'ea' not found. Please use a valid unit."
Those reflect incapacity to calculate the weight of ingredients such as for these 4 items:

Line 15: SPICE MASALA GARAM BULK	0.000 lb
Line 132: MISC GROCERY DRY GOODS	5.000 EA
Line 139: Macarons Signature Assorted 3/35ct	525.000 EA
Line 172: MISC GROCERY DRY GOODS	2.000 EA

One could ignore those as insubstantial contributors to total effects, or ask the vendor to clarify.

type02

1) If the food procurement worksheet is in a format like this, with total weight expressed in pounds but no column showing that unit of weight is pounds (e.g., FreshPoint's or US Foods)

Item Description	Item #	Weight	Qty Shipped	Total Weight
MILK WHOLE 4/1 GAL *RG*	77820	32	1	32
EGGS HARDBOILED WHL PLD 'TABLE READY'	85300	50	2	100
CHEESE QUESO FRESCO 5# *SO*	70674	5	1	5
MILK WHOLE GALLON *RG**EACH*	78606	8	2	16
APPLE GRANNY SMITH 80 CT	1228	40	1	40

Then first save the worksheet with a single header row renaming the "Total Weight" or "Net Wght Shipped" or other column reflecting the full product weight shipped to "product_weight", & "Item Description" or the like to "product" and add an empty column with a column name of "unit," rename a column with e.g, "Qty Shipped" to "qty" and remove any "Total" rows under the product listings. Then somewhere in the first column (e.g., in the row after the last product) include "type02" (lower case, no space). One might then run the Jupyter Notebook after specifying the name of the targeted procurement worksheet in the code near the top of the first cell.

type03

4) If the food procurement worksheet is in a format like this, with product weight information integrated into a product description field (e.g., Yale's Baldor worksheet):

Item Description	Stock Unit	Qty Sold CW
CHEDDAR CLOTHBOUND AGED CABOT 30 LB MC	LB	19
BLUE BAYLEY HAZEN BLUE 7 LB JASPER HILL	LB	3
SOUR CREAM TUB 5 LB CABOT	EACH	1
SMOKED SALMON SILVER LABEL SLCD WHOLE SIDE 3-4 LB CATSMO LCL	LB	23
YOGURT DRINKABLE BLACKBERRY 12X12 OZ RB	CTN	19
CANNOLI CREAM 6X2.5 LB GOLDEN CANNOLI	CTN	8

Change the "Item Description" column name to "product" and "Qty Sold CW" column name to "qty" (and, if need be, remove a bottom "Grand Total" line). Then somewhere in the first column (e.g., in the row after the last product) include "type03" (lower case, no space).

Please feel welcome to seek free assistance from stephen.ronan@wfa.org in prepping a worksheet in any different type of format.

ADDITIONAL NOTES

The process of the full Notebook involves 1) identifying ingredients of each product and their weights, 2) assigning those to categories and 3) calculating impacts, using the factors.csv file.

By default the code will go through that full 3-step process. But if one looks at the ingredients output file and decide to fix anything there, one could do so and then run the code again after, in the first Cell changing START_MODE from "full" to "after_ingredients".

Similarly, one could edit the categories output file and change START_MODE to "after_categories". Just remember to change that back before assessing another, new input file.

Another option that can be changed in the first Cell is this one, which defaults to 2.1%
CATEGORY_PERCENT_MIN = 2.1
That aims to eliminate, for example, additives that constitute a very small fraction of the product.

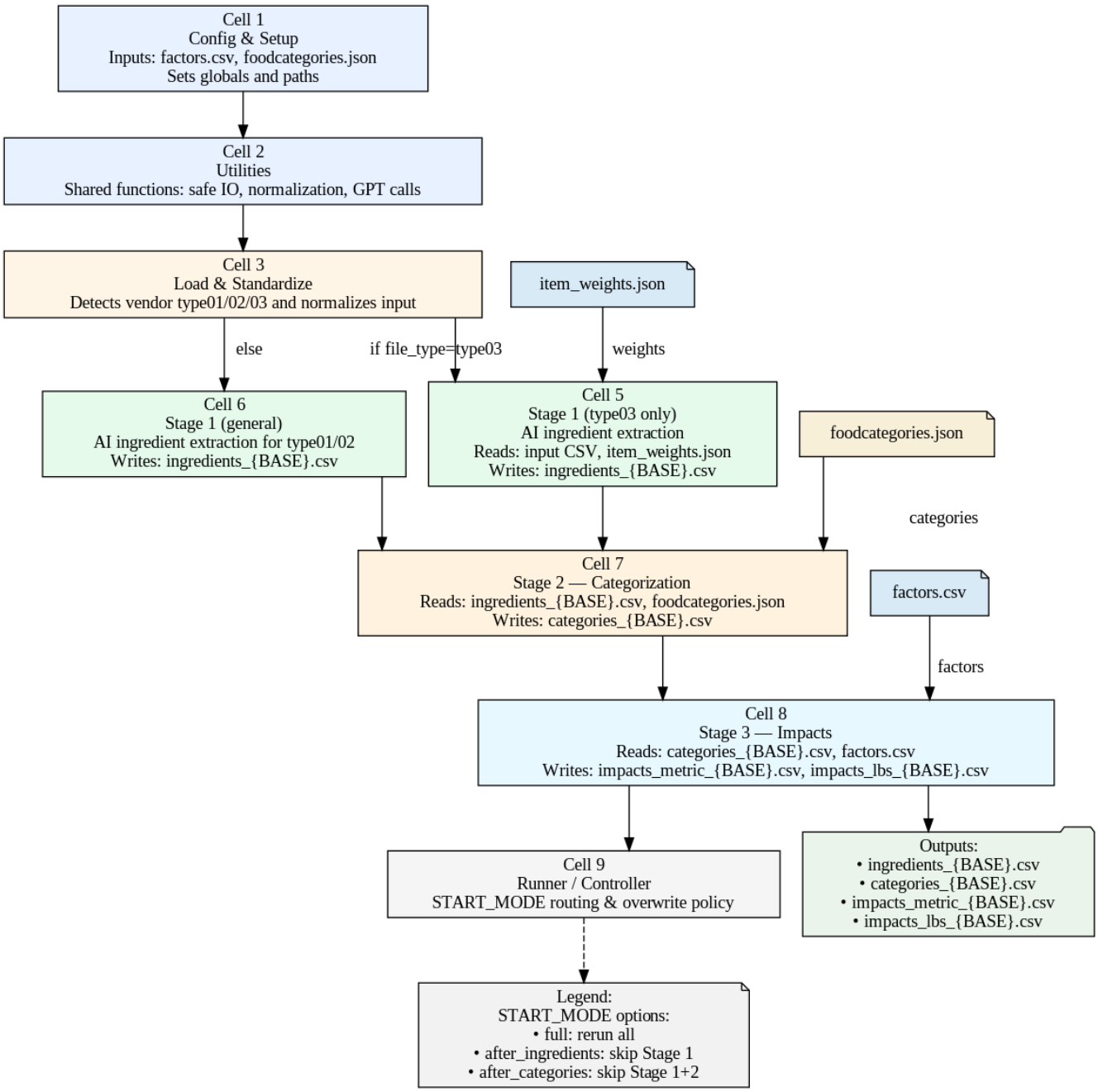
The Notebook can process a worksheet combining units of lbs, kilos, grams, gallons, liters etc.

Ingredient and category output files have an initial column whose numbers match the row number of the original product… So if a product has six ingredients there should be six rows that have the same number, which should match the row of the multi-ingredient product in the input file. The output may not all be in consecutive order because of the parallel processing, but one can sort the output file on that column to make it easier to compare the ingredient/category output with the original list of products.

Impact Files

The final output will be two .csv files written to disk, one with units in lbs/gallons/acres, its name including "impacts_lbs" as well as the input file name and the other with units in kilograms/liters/sqmeters, its name including: "impacts_metric" as well as the input file name.

Summary of the functions and input and output files of the Jupyter Notebook cells



FILES FOUND IN THIS FOLDER
https://drive.google.com/drive/folders/1HxQb4_QDudL_PlNLAQwHUMQrhZ54FzQV?usp=sharing

1) _AllFilesForDownload.zip
includes all the files below.

2) _documentation10.pdf
Describes the project and methods for assessing impacts of food procurement spreadsheets.

3) factors.csv
The conversion factors used to calculate the GHG emissions, cost in animals' lives, freshwater withdrawals, etc. from purchase of a given weight of a food product.

4) foodcategories.json
A .json file including Ingredient:Category combinations. The code first looks to see if an ingredient is included there and, if so, uses the category for the ingredient. If not, AI provides a category ingredient pair for immediate use and whose future fate depends on the setting for the FOODCAT_UPDATE_MODE variable in Cell 1
That defaults to "separate" but can be changed to "append" or "disabled
"append" - Adds new pairs directly to foodcategories.json.
"disabled" - Does not store new pairs anywhere (pure lookup mode)
"separate" - Adds new pairs to newcategories.json (It's recommended to review for appropriateness before merging into foodcategories.json

5) sortfoodcatmerge.ipynb
Can merge ingredient:category pairs in newcategories.json into the foodcategories.json file, sorting that updated foodcategories.json file by category, then ingredient name. The second cell in that sortfoodcatmerge.ipynb file could be run by itself to just sort foodcategories.json if you had been using the append method.

5) item_weights.json
Includes some typical weights for items such as lemons, green peppers, pineapples where input files identify quantity such as CS, CT or BSH (case, count, bushel) with total weight not provided.

5) FoodImpacts_v10d.ipynb
Jupyter Notebook code that works for most food procurement spreadsheets, once header row column names are assigned (e.g., product, product_weight, unit, qty)

6) usfoods.csv
Purchases by Yale University from a particular vendor... Used as a sample object for testing the system. Other such samples in a variety of formats available on request.

There are a variety of other files freely available that, for example, demonstrate in detail how impacts are calculated, e.g., the sources for determining the conversion file (factors.csv).

For copies of such additional files please contact stephen.ronan@wfa.org.

P.S. To experiment, we used a script to translate the 1500+ products in the US Foods worksheet into French, Spanish and German, with results like in the 2nd column below.

SPICE, PEPPER BLACK GROUND PLASTIC JUG 30M	Gewürz, Pfefferschwarz gemahlener Plastikkrug 30m
BEAN, NAVY DRIED RAW	Haricot, marin séché brut
PASTA, COUSCOUS ISRAELI SHELF STABLE IMPORTED	Pasta, establo israelí del cuscús importado
CABBAGE, KIMCHI SHRED PICKLED GLASS JAR REF	Kohl, Kimchi Shred eingelegtes Glasglas Ref
WATER, PURIFIED PLASTIC BOTTLE VARIABLE	Eau, variable de bouteille en plastique purifiée
PEANUT BUTTER, SS CUP	Mantequilla de maní, taza SS
PASTA, SPAGHETTI THIN SHELF STABLE	Pasta, Spaghetti dünne Regal stabil
PEPPER, CHIPOTLE SHELF STABLE	Poivre, étagère chipotle stable
JUICE BASE, APPLE 100% 5:1 DISPENSER CARTRIDGE FR	Base de jugo, manzana 100% 5: 1 Cartucho de dispensador congelado
RASPBERRY, RED PUREE FROZEN	Himbeere, rotes Püree gefroren
PINEAPPLE, CHUNK IQF FROZEN	Ananas, morceau iqf gelé

And then ran one of our Notebooks, unaltered, on the non-English product list, with less than 2% difference in result for each item in the total impacts row, comparing the run of code using the English and non-English versions of the product list.

P.P.S. If you need an OpenAI API key in order to run this, please write us at stephen.ronan@wfa.org